

COMP 6600 - Final Project Report

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Abstract

Accurate satellite orbit prediction underpins navigation, communication, and Earth observation. Traditional methods like the Simplified General Perturbation 4 (SGP4) model are efficient but accumulate error over time due to simplified force modeling and unmodeled perturbations. With the rise of precise tracking data, data-driven techniques now offer a way to enhance prediction accuracy.

This project develops a two-phase hybrid framework combining physics-based propagation and machine learning correction for the LAGEOS-1 satellite. Using Two-Line Element (TLE) data from CelesTrak and precise SP3 ephemerides from NASA Earthdata.

1. Introduction

The problem of accurately estimating satellite positions is critical in modern navigation, communication, and Earth observation systems. Satellite position information underpins applications such as Global Navigation Satellite Systems (GNSS), weather prediction, and satellite-based internet services. Traditionally, position estimation relies on physics-based orbital mechanics models or Kalman filtering techniques, which require detailed physical parameters and can be sensitive to modeling errors. With the growing availability of large-scale satellite tracking data, neural networks provide a promising alternative, capable of learning complex nonlinear relationships between orbital parameters and satellite positions.

1.1. Problem Statement

The core problem is to build a model that can predict satellite positions at future time steps, given past trajectory data or orbital parameters. Specifically, the dataset will consist of publicly available satellite ephemeris data. The expected result is a neural network model that can accurately estimate the position of satellites in Earth-Centered Earth-Fixed (ECEF) coordinates.

The method proposed is to develop and train a neural network architecture that can capture temporal dependencies in

satellite trajectories. While existing implementations for orbital propagation (such as SGP4 models) are widely used, the plan is to use them both as a source of labeled training data and as baselines for comparison. Improvement over these baseline models will be sought by leveraging the neural network's ability to learn nonlinear error patterns and generalize to different orbital regimes.

1.2. Problem Approach

The overall approach for this project is a two-phase machine learning framework designed to improve orbit prediction accuracy by learning the residual errors between physics-based and observed satellite trajectories. The project will utilize LAGEOS-1 satellite data, combining Two-Line Element (TLE) sets with precise orbit ephemerides. The goal is to train a machine learning model capable of forecasting and correcting the systematic propagation errors of the SGP4 model, thus improving the long-term prediction accuracy.

In the first phase, TLE and SP3 datasets will be processed to extract satellite positions and velocities over matching time intervals. The SGP4 propagator will generate predicted orbits from the TLEs, while the SP3 data will serve as ground truth. The difference between these datasets will form the training data for the machine learning model.

The second phase involves training and applying a time-delay neural network (TDNN) to model the error evolution over time. The network will use a convolutional layer to extract temporal features, followed by dense layers to forecast future error corrections. Once trained, the TDNN will predict SGP4 error trends and apply these corrections recursively to improve orbit forecasts.

2. Background and Related Work

2.1. Satellites Orbit Information

2.1.1. GNSS Orbit Propagation

Global Navigation Satellite Systems (GNSS), including GPS (United States), Galileo (European Union), GLONASS (Russian Federation), and BeiDou (China), rely

on precise satellite position and timing information to enable accurate navigation. Since users on Earth cannot measure satellite orbits directly, each GNSS satellite broadcasts its predicted orbital data, known as the ephemeris, which allows receivers to compute the satellite's location in space at any given time.

Ephemeris refers to a model of a satellite's future position, velocity, and clock behavior. It is generated continuously by ground control stations that monitor satellites using a network of high-precision tracking antennas. Using orbit determination algorithms (e.g., batch least squares, Kalman filtering), these stations estimate the satellite's state vector and clock drift, along with corrections for forces such as Earth's gravity field, solar radiation pressure, and tidal effects. The resulting predictions are uploaded to each satellite and broadcast as part of the GNSS navigation message. GNSS ephemeris propagation is shown in 1.

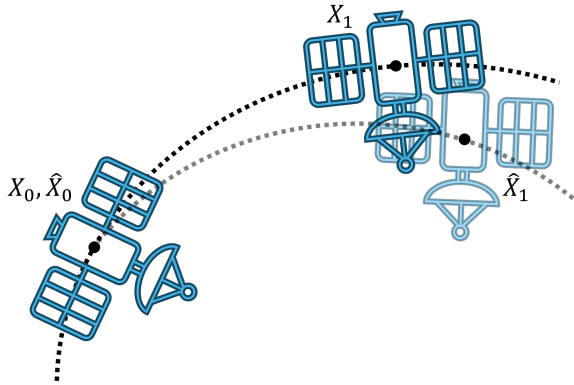


Figure 1. GNSS Propagation Error Visual

GNSS ephemerides are typically valid for a few hours and are optimized for real-time navigation, not long-term orbit prediction.

2.1.2. LEO Orbit Propagation

Orbital information for Low Earth satellites is typically represented using the Two-Line Element (TLE) format, which encodes mean orbital elements derived from radar and optical tracking by the U.S. Space Surveillance Network (SSN). TLEs are publicly distributed by the U.S. Strategic Command and are widely used due to their accessibility and standardized structure. Each TLE represents a mean Keplerian state corrected for perturbative effects such as Earth's oblateness and atmospheric drag. These keplerian states are propagated by a Simplified General Perturbations 4 (SGP4) model to update a satellite's position and velocity.

However, while TLEs provide a convenient global catalog, they are not inherently precise. Their accuracy depends on the underlying fitting model, observation density,

and update frequency. Studies such as Wei and Zhao [2] and Aida and Kirschner [1] have demonstrated that TLE-based position estimates can deviate by hundreds of meters to several kilometers, depending on altitude and propagation time. These inaccuracies make raw TLE data inconsistent for applications requiring high-precision orbital prediction, such as collision avoidance and autonomous navigation, shown in 2.

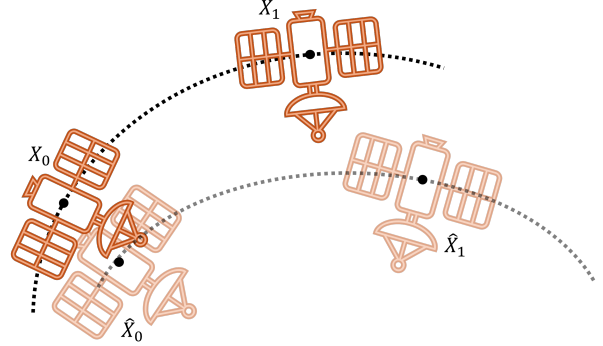


Figure 2. LEO Propagation Error Visual

LEO satellites may also broadcast ephemeris to users, similar to GNSS constellations. While these ephemeris are more accurate than SGP4, the accuracy is not guaranteed to be at the same level as that of its GNSS counterpart. Additionally the ephemeris is only able to be propagated over a predetermined window, any propagation outside of that time frame results in large magnitude of position error.

2.2. Orbit Propagators

Orbit propagation models transform an initial orbital state forward in time to estimate future positions and velocities. These propagators generally fall into two categories: numerical and analytical.

Numerical propagators, such as the High-Precision Orbit Propagator (HPOP), integrate the equations of motion using detailed physical models that include high-order gravity terms, solar radiation pressure, atmospheric drag, and third-body perturbations. They achieve centimeter- to meter-level accuracy when properly initialized but are computationally intensive and require detailed spacecraft parameters that are often unavailable for public datasets.

Analytical propagators, such as the Simplified General Perturbation models (SGP4/SDP4), provide a computationally efficient alternative. They approximate orbital motion using mean element models and are the standard for propagating TLEs. However, their simplified physics introduces cumulative errors over time.

Extensive evaluations have quantified SGP4's accuracy across orbital regimes. Wei and Zhao [2] analyzed 1,120

space objects and reported orbit determination errors on the order of 100 m for low-Earth orbits (LEO), 0.7 km for medium-Earth orbits (MEO), and 1.9 km for geosynchronous (GEO) satellites, with typical prediction errors remaining under 40 km for up to 30-day propagations. Aida and Kirschner [1] compared SGP4 outputs with GPS-based reference orbits from GRACE-1, TerraSAR-X, and TET satellites, finding that along-track errors dominate and grow approximately quadratically with time, reaching tens of kilometers after a week. These results underscore SGP4's efficiency but also its inherent limitations in precision and long-term reliability.

2.3. Use of Deep Learning in Orbit Propagation

With the growing availability of precise ephemerides from missions and commercial constellations, data-driven methods have emerged as powerful tools for orbit prediction. Traditional physics-based models can struggle with unmodeled perturbations, atmospheric variability, and non-conservative forces, whereas machine learning (ML) techniques can approximate complex nonlinear relationships directly from observed data. These approaches are especially relevant in low-Earth orbit (LEO), where drag, solar activity, and density fluctuations introduce highly nonlinear effects that are difficult to represent with simplified analytical propagators such as SGP4.

Research in this field has primarily followed two main strategies: direct orbital prediction and residual error correction. Although both rely on learning temporal dynamics from data, they differ in scope, data representation, and their reliance on physical models.

In the direct orbital prediction approach, a neural network is trained to forecast future satellite states typically position and velocity vectors directly from past orbital data. This method treats orbit propagation as a sequence forecasting problem, relying entirely on data to learn the underlying motion. The key advantage is flexibility: the network can, in principle, learn complex environmental dependencies without requiring explicit modeling of physical forces. However, this also introduces drawbacks. Because no physical constraints are imposed, prediction errors can accumulate rapidly, leading to drift or instability during long-term propagation. Models trained in this manner also tend to generalize poorly when exposed to conditions or orbital regimes not represented in the training set.

The second strategy, known as residual error correction, combines the strengths of analytical propagation and machine learning. In this hybrid framework, a baseline physics-based propagator such as SGP4 provides the nominal orbit prediction, while an ML model is trained to learn the residual discrepancy between this analytical output and high-accuracy reference data (for instance, from numerical propagators or precise ephemerides). This approach ef-

fectively offloads the burden of modeling well-understood dynamics to the analytical propagator, allowing the ML model to focus on compensating for unmodeled perturbations such as drag variability, solar flux changes, and higher-order gravitational effects. Because the network operates on smaller residual magnitudes rather than absolute position values, it generally requires less data and converges more efficiently. Importantly, this formulation preserves the physical interpretability of the baseline model while improving its overall accuracy.

A representative example of this hybrid approach is presented by Kassis et al. [4], who trained neural networks to map SGP4 outputs to high-fidelity trajectories generated by the High-Precision Orbit Propagator (HPOP) for Orbcomm satellites. By learning the residual error between SGP4 and HPOP, their *SGP4 + ML* framework achieved sub-10-meter position accuracy while maintaining real-time computational efficiency.

2.4. Time-Dependent Neural Networks

A wide range of neural network architectures have been investigated for orbit prediction and estimation, each with varying levels of success depending on the data structure and temporal complexity of the problem. The IPA Machine Learning for Orbit Determination study [3] and the Kassis study [4] both compared architectures such as feedforward neural networks (FNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs) to assess their suitability for learning satellite motion dynamics. These studies consistently found that models capable of representing temporal dependencies outperform static networks that process each orbital state independently. This is because orbital motion, especially in low-Earth orbit (LEO), exhibits strong time-correlated behavior due to periodic gravitational, atmospheric, and solar effects.

Among the time-dependent architectures tested, the Time-Delay Neural Network (TDNN) and the Long Short-Term Memory (LSTM) network have emerged as particularly effective for orbit prediction tasks. Both architectures are designed to capture sequential dependencies in time-series data, but they differ in how they represent and retain temporal information. The TDNN extends a traditional feedforward neural network by incorporating delayed versions of the input signal, allowing the network to learn temporal correlations over a fixed window of previous states. This makes the TDNN especially well-suited for short-term orbital forecasting, residual correction, and real-time applications where the model must operate efficiently with limited computational resources. Kassis demonstrated that TDNN-based hybrid models, when used to learn residual errors between SGP4 and high-fidelity HPOP trajectories, significantly reduced propagation error while maintaining real-time performance [4].

The LSTM, by contrast, is a recurrent architecture that employs gated memory cells to selectively retain or forget information over longer sequences. This capability enables LSTMs to capture complex, long-term dependencies in orbital behavior, such as gradual drift due to atmospheric drag or solar activity cycles. The IPA study showed that LSTMs generalize better than feedforward or convolutional models when trained on noisy or sparse orbital datasets, as their recursive design naturally models temporal continuity and error accumulation. LSTMs are therefore particularly beneficial for predicting long-duration orbits and learning slow-varying perturbations that cannot be captured effectively by shallow or static models.

3. Project Overview

The overall approach for this project is a two-phase machine learning framework designed to improve orbit prediction accuracy by learning the residual errors between physics-based and observed satellite trajectories. Specifically, the project will utilize LAGEOS-1 satellite data, combining Two-Line Element (TLE) sets from CelesTrak with precise orbit ephemerides in SP3 format obtained from NASA’s Crustal Dynamics Data Information System (CD-DIS), a database that archives space geodesy data. The goal is to train a machine learning model capable of forecasting and correcting the systematic propagation errors of the SGP4 model, thus improving the long-term prediction accuracy.

The first phase of the framework focuses on the data collection and preparation. The two datasets, TLE and SP3, will be processed to extract the corresponding satellite positions and velocities over identical time intervals. The SGP4 propagator will be used to compute positions from the TLEs, while the SP3 data will serve as the truth reference. The difference between these two data sources will form the primary input to the learning model. The error that will be used as the training data can be seen in Figure 3. This “error-based” training strategy is critical because it allows the model to learn the discrepancies between the well-understood propagators and truth reference, rather than directly modeling the orbital dynamics themselves. This is beneficial because the orbital mechanics are governed by complex and nonlinear perturbative forces. This approach effectively isolates the behavioral differences between the analytical and observed trajectories, reducing the degree of nonlinearity the network must learn.

The second phase involves developing and training the machine learning model. The TDNN used in this project, shown in Figure 4, is composed of a sequence of dilated one-dimensional convolutional layers that operate along the temporal axis of the orbit error sequences. Each layer applies left-sided causal padding so that the convolutional filter only processes the current and past error samples. This

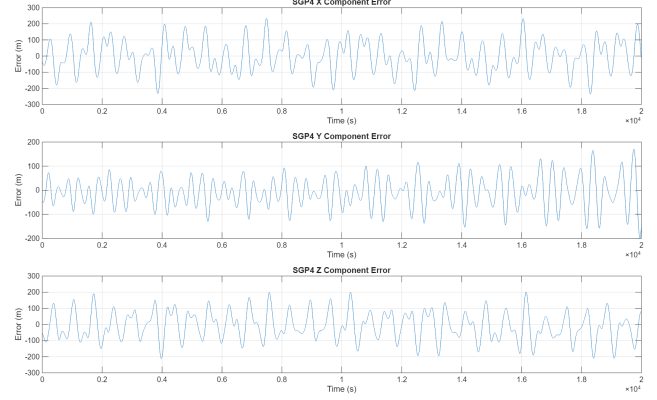


Figure 3. Component-wise position errors (X, Y, Z) between SGP4 and SP3 orbits in the ECI frame.

ensures that the network adheres to real-time prediction constraints and does not access future information. After each convolution, a normalization layer and nonlinear activation function (ReLU) are applied to stabilize the layer outputs and improve learning efficiency. Residual connections between TDNN blocks are incorporated to preserve information flow and mitigate vanishing-gradient effects. After the convolutional layers, a global pooling layer compresses the feature maps into a fixed-length representation. That is then passed through fully connected layers that generate the final predicted error corrections in the ECI frame.

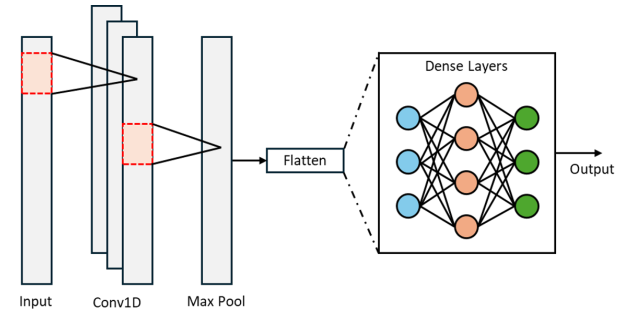


Figure 4. TDNN architecture diagram.

Once trained, the TDNN will be used to forecast SGP4 propagation errors and apply corrections to future orbit predictions. During this operational phase, the latest TLE and SP3 data will be retrieved, and the network will use the most recent error samples to predict future correction terms. The corrected position and velocity estimates will then be obtained by adding the model’s predicted errors to the SGP4-propagated outputs. The training and testing process is shown in Figure 5.

The performance of the framework will be evaluated through trajectory comparison plots and statistical error metrics such as Root Mean Square Error (RMSE). Improve-

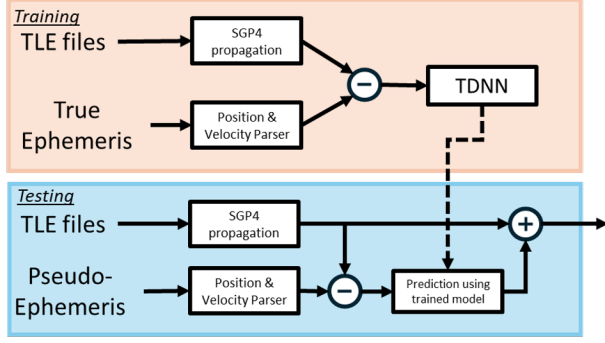


Figure 5. Architecture and information flow of the proposed framework.

ments over baseline SGP4 propagation will demonstrate the capability of the neural model to generalize across time and provide enhanced prediction fidelity.

4. Code Reuse

This project relied primarily on original code written specifically for the development of the Time Delay Neural Network, the construction of windowed residual datasets, and the evaluation of SGP4 correction performance. Several standard Python libraries were used without modification, including NumPy, Pandas, PyTorch, and Matplotlib, which provided numerical operations, data handling, neural network components, and visualization tools. The general structure of the PyTorch training loop, including the use of the Adam optimizer and mean squared error loss, follows conventions found in the official PyTorch tutorials, but no tutorial code was directly copied or reused.

The MATLAB preprocessing code base consisted of a larger set of scripts used to parse, propagate, align, and compare the LAGEOS-1 orbit data. Several MATLAB functions were reused from existing toolboxes and external utilities. The function `tlread` originated from the Aerospace Toolbox and was used to convert TLE files into MATLAB TLE objects. The propagation of each TLE was performed using `propagateOrbit`, also provided through MATLAB's Aerospace Toolbox, which implements the SGP4 algorithm. SP3 files were read using the function `readSP3`, a file parser created to obtain position, velocity, and time. Other functions such as `parseMultiTLE`, were written to be able to use SP3 and TLE files effectively. All higher-level logic in the MATLAB script and the Python TDNN model, including the looping over SP3 files, matching each SP3 segment to the appropriate TLE epoch, converting coordinates, computing component-wise residuals, and generating the error tables, were written specifically for this project.

5. Experimental Setup

The experiments conducted in this project were designed to evaluate whether a Time-Delay Neural Network (TDNN) can effectively learn and forecast the systematic propagation errors produced by the SGP4 orbit model when compared to precise SP3 ephemerides. The overall objective was to determine whether error-based learning could capture the temporal structure of these residuals and whether the resulting model could generate future corrections that improve orbit prediction accuracy. All experiments were performed using synchronized pairs of LAGEOS-1 TLEs from CelesTrak and SP3 precise ephemerides from NASA CDDIS, with the SP3 data serving as the truth reference. After preprocessing, the difference between SGP4-propagated and SP3-observed positions formed the time-series dataset used to train the TDNN seen previously in Figure 3. These residuals were then segmented into fixed-length temporal windows of 180 samples, each paired with the ground-truth next residual. The input features included the residual history, its first difference, and normalized SGP4 position and velocity components, reflecting the full state history available to the forecasting model.

The primary experiment involved training the causal TDNN to predict the next-step error using this windowed dataset and then evaluating the model on previously unseen files held out during training. The TDNN was trained using mean squared error (MSE) loss and the Adam optimizer. The learning process was monitored using training and validation loss curves, which consistently decreased and diverged minimally, indicating stable training dynamics and limited overfitting. After training, the model was applied to the held-out test sequences, shown in Figure 6, to generate predicted residuals, which were then added to the raw SGP4 positions to obtain corrected orbital trajectories.

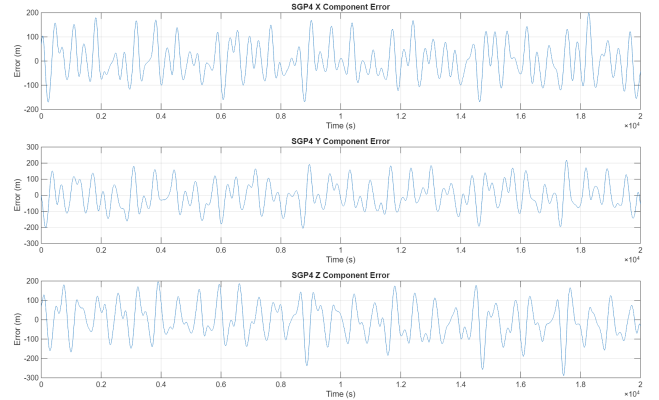


Figure 6. Testing Data Set: Component-wise position errors (X, Y, Z) between SGP4 and SP3 orbits in the ECI frame.

Performance was quantified using multiple evaluation metrics, including the 3D Euclidean error between the

corrected trajectory and the SP3 truth, as well as Root Mean Square Error (RMS) comparisons between SGP4 and TDNN-corrected predictions. The TDNN demonstrated measurable improvements, reducing the RMS error relative to baseline SGP4 propagation, shown in Table 1.

Model	Position RMSE (m)
SGP4	153.77
TDNN	83.95
%45.40 Improvement	

Table 1. SGP4 vs TDNN

In addition to quantitative metrics, several qualitative analyses were performed to visualize the model’s behavior. Residual prediction plots, shown in Figure 7, displayed that the TDNN closely tracked the true SP3–SGP4 error signal across all three ECI components, capturing both the slow drift and the finer oscillatory patterns characteristic of orbital error growth. The predicted curves followed the true residuals not only in overall magnitude but also in timing, phase, and sign changes, demonstrating that the model learned the underlying temporal structure rather than simply smoothing the signal.

Overlay plots of SGP4 error versus TDNN-corrected error, seen in figure 8, further illustrated that the corrected trajectories consistently remained below the baseline error for extended periods, with the largest reductions occurring during regions where SGP4 normally accumulates long-term drift.

Scatter plots comparing the true and predicted residual components, seen in figure 9, showed tightly clustered point clouds along the one-to-one diagonal line, indicating strong agreement and minimal bias in each spatial dimension. It is important to note that there is a strong bias in each direction of the components.

Error histograms, shown in figure 10, revealed that the TDNN both narrowed the distribution and shifted it closer to zero, reducing large outliers and producing a more concentrated and predictable error profile. Together, these visual and numerical results confirm that the TDNN successfully learns the temporal dynamics of orbital propagation errors and provides a meaningful improvement in prediction accuracy when applied to new data.

6. Conclusion

This project demonstrated that a Time-Delay Neural Network can successfully learn the temporal structure of SGP4 propagation errors and provide meaningful next-step residual predictions that improve orbit accuracy when applied to a satellite. By training on windowed sequences of historical SGP4–SP3 residuals, the TDNN learned to capture both short- and long-range dependencies, including the os-

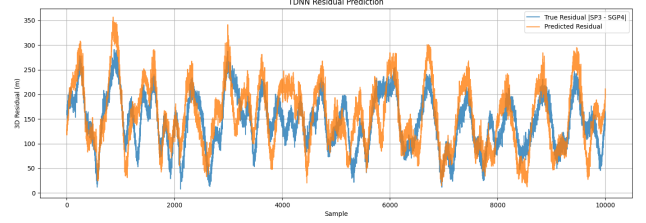


Figure 7. Error Residual Comparison: SGP4 residual error and the model predicted residual.

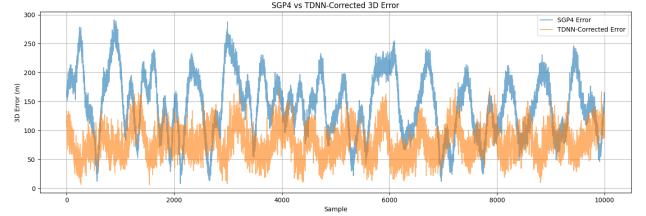


Figure 8. Corrected 3D Error Comparison: SGP4 3D error and the model corrected 3D error.

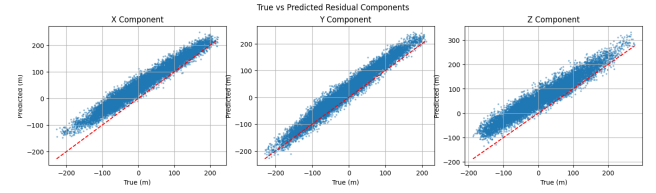


Figure 9. Residual Component Comparison: Component-wise residual errors (X, Y, Z) between true orbit and TDNN corrected errors.

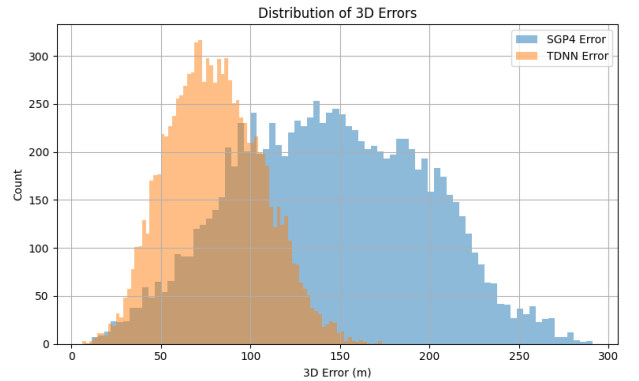


Figure 10. Distribution of Error Comparison: SGP4 error distribution and TDNN corrected error.

cillatory signatures characteristic of orbital perturbations. The experimental results showed measurable quantitative improvement in 3D position accuracy on held-out data, as well as strong qualitative alignment between predicted and

true residuals. These findings validate the idea that residual learning offers an effective way to enhance analytical orbit models without requiring modification to underlying physics models or access to detailed force modeling.

Throughout the project, several learning objectives emerged. One major lesson was determining how to structure training/validation/testing datasets. Designing an appropriate split required careful consideration of the temporal nature of the data and the risk of information leakage. To address this, entire files were grouped together and assigned to either the training or the testing set, ensuring that the model was evaluated only on completely unseen time periods. The project also provided valuable experience with debugging complex systems. Identifying issues related to data alignment, inconsistent scaling, and architectural behavior required careful inspection of intermediate outputs. This reinforced the importance of validating each step in a data pipeline when troubleshooting interdependent components.

Several meaningful directions can extend this work and further develop the approach explored in this project. One potential direction is to evaluate more complex neural network architectures that may better capture the long duration temporal patterns present in orbit propagation errors. Examples include deeper convolutional models with larger receptive fields, temporal convolutional networks, recurrent neural networks such as LSTMs and GRUs, and attention based models that can analyze long sequences of error history. These alternatives may provide improved handling of complex relationships within the residual signals. Another promising idea is to enhance the input feature set with additional physical and environmental information. Features such as solar flux measurements, geomagnetic activity indices, atmospheric density estimates, or TLE quality metrics could help the model account for perturbations that are not fully represented in the SGP4 analytical model.

7. Division of Labor

A detailed breakdown of individual contributions is provided below:

- Oren Miller worked on the data preparation and was also responsible for working on the machine learning pipeline used. In data preparation, Oren implemented the preprocessing code to synchronize TLE and SP3 epochs to align all data for each of the files. His work also included constructing the Time-Delay Neural Network (TDNN), data loading architecture, and implementing the network hyperparameters.
- Luke Zins also contributed to both the data preparation components of the project and the Machine learning model. He retrieved and organized the SP3 precise ephemerides for LAGEOS-1 from NASA's CDDIS. He also retrieved the TLEs from CelesTrak. He worked on

the preprocessing code to execute the SGP4 orbital propagation, and generated the error tables used for model training. He also performed validation to ensure correct frame alignment, time matching, and error component extraction. Luke also helped debug the initial machine learning code base until a working model ran.

After the initial development phases, both members collaboratively refined the end-to-end system improving the model. This included debugging inconsistencies in the training data, adjusting multiple parameters, and validating the performance of the corrected orbit predictions. Both members contributed to the problem formulation, literature review, creation of figures, and final document preparation.

References

- [1] Saika Aida and Michael Kirschner. Accuracy assessment of SGP4 orbit information conversion into osculating elements. In *Proceedings of the 23rd AAS/AIAA Space Flight Mechanics Meeting*, Kauai, HI, 2013. 2, 3, 7
- [2] Wei Dong and Chang-yin Zhao. An accuracy analysis of the SGP4/SDP4 model. *Chinese Astronomy and Astrophysics*, 34(4):365–372, 2010. 2
- [3] Institute of Physics and Astronomy (IPA). Machine learning for orbit determination. In *International Conference on Space Operations and Dynamics*, 2022. 3
- [4] Sharbel Kozhaya, Jamil Haidar-Ahmad, Ali Abdallah, Zaher Kassas, and Samer Saab. Comparison of neural network architectures for simultaneous tracking and navigation with LEO satellites. In *Proceedings of the ION GNSS+ Conference*, pages 1–10, St. Louis, MO, 2021. 3
- [5] Yang Yang, Michael Kirschner, and Saika Aida. Reliability analysis of NORAD TLE using GOCE satellite. In *AIAA/AAS Astrodynamics Specialist Conference*, San Diego, CA, 2014. 7

[1–5]